Sparse Autoencoders for Mechanistic Interpretability in NLP: A Critical Survey of Recoverability, Interpretability, and Utility

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Abstract

Sparse autoencoders (SAEs) have emerged as a promising approach to mechanistic interpretability by decomposing language-model activations into sparse, potentially interpretable features. However, differences in design assumptions, evaluation objectives, and experimental settings complicate conclusions about their effectiveness. We present a critical survey of 138 studies on SAEs in natural language processing. Our contributions are fourfold: (1) we propose a task-grounded taxonomy connecting dictionary design, feature operations, and NLP functions; (2) we examine how sparsity allocation, geometry, data coverage, and inference affect feature recoverability; (3) we assess evidence for semantic interpretability, mechanistic faithfulness, and behavioral utility; and (4) we synthesize findings across multilingual processing, retrieval, reasoning, classification, and behavioral regulation, highlighting the roles of objectives, baselines, supervision, and intervention pipelines. Through comparative analyses of published results, we reconcile apparent disagreements, identify unresolved questions, and propose directions for more rigorous evaluation. An anonymous research register[1] accompanies the survey.

1 Introduction

Sparse autoencoders encode model activations as sparse coefficients over a learned dictionary of feature directions (Cunningham et al., 2024). They make it possible to inspect concepts, compare representations, and intervene on internal variables at a finer granularity than whole layers or attention heads. Sparse interpretable sentence embeddings provide an earlier precedent (Trifonov et al., 2018); transformer studies established useful feature decompositions (Bricken et al., 2023; Cunningham et al., 2024); subsequent work scaled dictionaries and released reusable resources (Templeton et al., 2024; Gao et al., 2025; He et al., 2024b). The scientific opportunity is now broader: can these coordinates explain how language is represented and processed, and can that understanding improve NLP systems?

Answering this question requires separating four outcomes. *Recoverability* concerns whether an encoding procedure recovers the target latent structure, where that structure is specified. *Semantic interpretability* concerns whether a feature description makes discriminating predictions about activations or effects. *Mechanistic faithfulness* concerns whether an explanation identifies computations the original model uses. *Behavioral utility* concerns success on a stated task with its competing objectives. These outcomes can agree, but their definitions do not imply agreement.

Consider language identity. Multilingual feature analyses use it to distinguish language-associated representations (Deng et al., 2025); language steering seeks to induce a chosen output language (Gurgurov et al., 2026); LangSAE removes it from pooled embeddings to improve multilingual retrieval (Kim et al., 2026). Grouping these studies simply as multilingual SAE applications hides an objective-dependent reversal: information useful for generation can be a nuisance for semantic retrieval. Conversely, grouping only by encoder architecture does not reveal this distinction at all.

Existing surveys provide complementary perspectives on SAEs and mechanistic interpretability. Shu et al. (2025) synthesize SAE architectures, training procedures, feature explanations, evaluation methods, and applications. Mueller et al. (2026a) organize NLP mechanistic interpretability around causal mediators and search methods, examining which computational units and analytical procedures support explanation, hypothesis testing, and model editing. Zhang et al. (2026a) connect localization and steering to improvements in align-

[1] https://anonymous.4open.science/r/sae-research-register-C331/register.html

ment, capability, and efficiency, including SAE features, task-specific metrics, and checks for unintended effects. Building on these foundations, our survey examines how SAE construction and use jointly shape the comparability and interpretation of NLP findings.

This motivates three research questions. **RQ1: Which design assumptions change what an SAE can recover?** We compare allocation, geometry, data coverage, and inference under their stated objectives and theoretical assumptions (Section 3). **RQ2: What evidence distinguishes an interpretable or useful feature from a faithful mechanism?** We examine semantic tests, intervention controls, and counterexamples using a claim-specific rubric (Section 4). **RQ3: Why do NLP studies reach different conclusions about SAE usefulness?** We compare task objectives, supervision, and control pipelines, and reanalyse selected within-study results (Section 5).

Our taxonomy organizes SAE studies by dictionary design, feature operation, and NLP function. Our multi-label register and comparative analyses support a **task-grounded synthesis** that clarifies apparent disagreements and identifies unresolved questions.

2 Scope and Organizing Framework

We survey SAE methods for language-model analysis and NLP system construction, including relevant speech and multimodal extensions. Corpus selection and coding are detailed in Appendix A.

Three facets and their qualifiers. For study s, we record $T(s) = (D_s, O_s, F_s; Z_s)$ (Figure 1). **Dictionary design** D_s describes how features are obtained or constrained: sparsity allocation, geometry, coverage and objectives, inference and activation sites, or functional alternatives. **Feature operation** O_s describes their use: characterize representations, predict or diagnose outcomes, trace dependencies, or edit model behavior and construct systems. NLP function F_s specifies the evaluation target, covering linguistic distinctions, knowledge and retrieval, reasoning and adaptation, behavioral regulation, text analytics, speech and multimodal analysis, and general enabling research.

The facets interact: a task can shape dictionary training, while dictionary structure can enable particular operations. Qualifiers Z_s retain model/version, activation site, data, supervision, feature selection, and intervention protocol. Together, the upper boxes and central band in Figure 1 expose differences in experimental settings that may explain apparent disagreements.

Claim-dependent evidence. The taxonomy identifies comparable studies; the evidence framework assesses their claims. In the lower row of Figure 1, claims about feature recoverability, semantic interpretability, mechanistic faithfulness, and behavioral utility require different tests and controls. In particular, demonstrating the utility of SAE-derived features in a classifier, retriever, or controller does not necessarily establish the computations performed by the original model.

The operational definitions, study assignments, and cross-dimensional analyses are provided in Appendix B.

3 Dictionary Design and Recoverability

RQ1 asks which design assumptions determine what an SAE can recover. For an NLP researcher seeking a grammatical distinction, an unsuccessful feature search leaves several explanations open: the chosen representation, the dictionary's constraints, the training data, or the encoding procedure. We compare methods by which of these limitations they address.

For activation $h \in \mathbb{R}^d$, an encoder produces $z = f_\theta(h) \in \mathbb{R}^m$ and a decoder reconstructs $\hat{h} = b + Dz$. A common objective is

$$\mathcal{L} = \mathbb{E}[\|h - b - Df_\theta(h)\|_2^2 + \lambda\Omega(z)]. \quad (1)$$

This objective rewards reconstruction and sparsity; recovering a specified latent structure is a separate endpoint. Synthetic studies can compare against a known generating dictionary, whereas language-model studies must validate the proposed feature interpretation.

3.1 Defining the structure to be recovered

The activation site determines the starting object. Residual streams contain accumulated information; attention and MLP outputs characterize component contributions. Transcoders instead approximate a component's input–output map (Dunefsky et al., 2024; Paulo et al., 2025). Layer transfer, shared dictionaries, and routing change training cost and parameter reuse (Ghilardi et al., 2024, 2025; Shi et al., 2025); comparisons still need to establish what structure is preserved across sites.

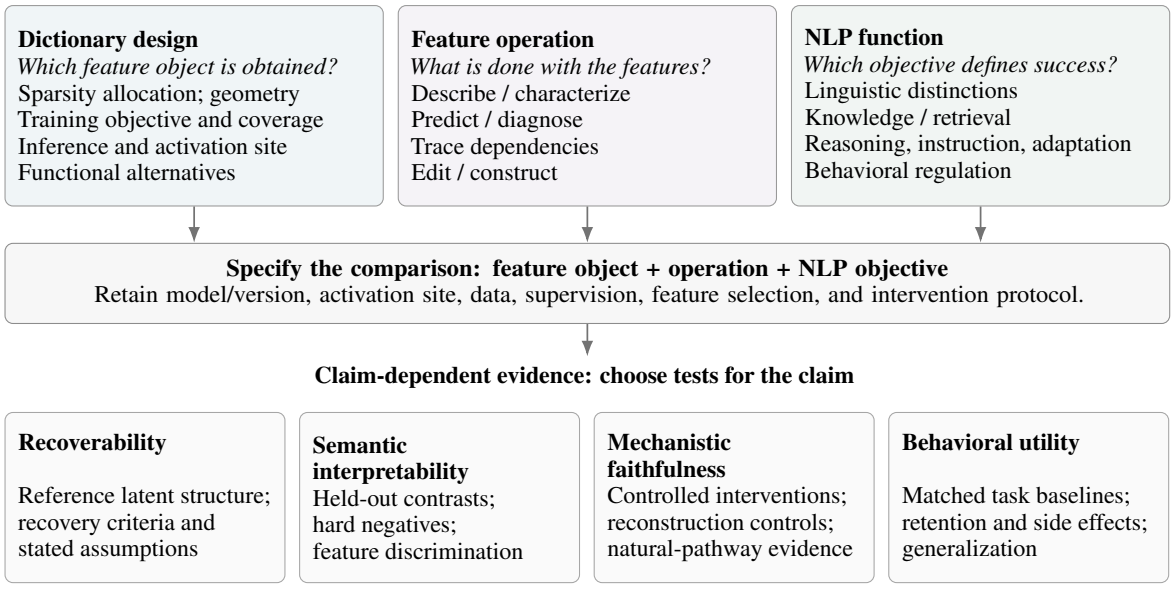


Figure 1: Three facets jointly define a comparison (top), interpreted alongside its experimental settings (middle). Complementary evidence targets assess what the findings support (bottom). Arrows show how the facets inform study comparison and claim assessment; the tests are illustrative and tailored to the claim.

Within a site, the question is how that structure should be represented. Splitting distributes a concept across latents, while absorption allows a specialized latent to carry a broader property (Chanin et al., 2025); neither supports equating an inactive coordinate with absent information. Subspace (Engels et al., 2025; Dalili and Mahdavi, 2026) and manifold approaches (Bhalla et al., 2026a) model structure beyond individual directions. Nested scales (Bussmann et al., 2025), parent–child gates (Muchane et al., 2025), and Kronecker factorization (Kurochkin et al., 2025) impose different forms of organization across features. Temporal smoothness (Bhalla et al., 2026b) and predictable-context/novelty separation (Lubana et al., 2026) instead organize variation over sequences. These choices change the candidate unit of analysis: their NLP value depends on whether the resulting directions, groups, or temporal features capture the distinction under study.

3.2 Allocating sparse capacity

Once the target is specified, allocation determines which features can participate in each reconstruction. An ℓ_1 penalty encourages sparsity but also shrinks amplitudes. Gated SAEs separate selection from magnitude estimation, while JumpReLU learns activation thresholds (Rajamanoharan et al., 2024a,b). TopK fixes the number of retained coordinates per input (Gao et al., 2025); BatchTopK distributes a budget across a batch and estimates an inference-time threshold (Bussmann et al., 2024). Top-AFA selects counts through a quasi-orthogonality approximation (Lee et al., 2025), AdaptiveK uses a learned complexity signal (Yao et al., 2026), and Group Bias Adaptation (GBA) targets activation-frequency coverage (Chen et al., 2026b). These allocation rules make different assumptions about inputs, geometry, and feature frequency.

The results show why allocation cannot be ranked by reconstruction alone. BatchTopK improves reconstruction over TopK, but its Gemma-2-2B comparison with JumpReLU is mixed for downstream cross-entropy degradation (Bussmann et al., 2024). In GBA's variable-sparsity synthetic experiment, TopK recovery falls from 98.4% at $K = 20$ to 23.4% at $K = 50$; full versus misspecified frequency coverage gives GBA 100% versus 3.9% recovery (Chen et al., 2026b). These rates use alignment threshold 0.9 and 128 generating features. They demonstrate sensitivity to allocation and coverage assumptions under known ground truth, rather than an NLP ranking. Comparisons should match average sparsity, data, and insertion protocol; decoder normalization must also be recorded because it changes coefficient scales (Appendix C).

3.3 Separating coverage from inference error

Even a suitable representation and budget cannot resolve every failure. Specialized sampling and tilted

risk improve access to rare concepts (Muhamed et al., 2025b), changing the training distribution and emphasis of the objective. Weighted SAEs analyze recovery and reweighting under an idealized tied-weight model (Cui et al., 2026); those assumptions delimit the result's applicability to language-model activations. Locally amortized inference addresses a different limitation: a shared encoder can leave an optimization gap relative to per-input sparse coding (Sun et al., 2026). Improving that approximation does not establish that the reference codes are the intended linguistic features. Thus data coverage, the learning objective, and encoding accuracy require separate comparisons.

Answer to RQ1. Recoverability has separable bottlenecks: the representation and its geometry, sparse allocation, training coverage, and inference. The evidence supports diagnosing the suspected bottleneck while matching the remaining conditions. Reconstruction, recovery against a specified reference, and behavioral preservation should remain separate endpoints. Technical comparisons and assumptions are detailed in Appendix C.

4 From Features to Explanations and Mechanisms

Recovering sparse features leaves their explanatory role unresolved. RQ2 asks what distinguishes a meaningful or useful feature from a faithful account of model computation. A feature may predict sentiment, support sentiment steering, or mediate the model's own sentiment decision; these claims require different tests. Table 1 organizes support for a specified claim and evaluation setting. Appendix D develops the assessment rules through worked claims and a causal counterexample.

4.1 Testing the feature's meaning

A feature description is a hypothesis about its behavior. Input-oriented descriptions predict what elicits activation; output-oriented descriptions predict contributions to generation (Gur-Arieh et al., 2025). Automated explanation (Paulo et al., 2024), agentic testing (Han et al., 2026), and counterexample-guided revision (Ma et al., 2025) make these hypotheses testable on held-out positives and hard negatives. The critical question is whether the description discriminates the intended property from alternatives. Descriptive collision tests whether one explanation also fits unrelated features (McCann, 2026); contrastive stories, semantic/functional neighborhoods, and concept-separability tests examine complementary aspects of specificity (Gulko et al., 2025; Mercorio et al., 2025; Fereidouni et al., 2026; Härle et al., 2025).

The controls must preserve what the claim says should remain invariant. A grammatical feature should track the specified contrast across appropriate lexical controls; a lexical feature need not survive word substitution. Adversarial inputs test the limits of a description (Li et al., 2026a), while randomized SAE components and randomized host models challenge different sources of apparent interpretability (Korznikov et al., 2026; Heap et al., 2026). In the latter comparison, similar aggregate scores in several tested settings show that those metrics need not distinguish learned computations from structure present without training (Heap et al., 2026). Cross-run consistency asks whether features recur across learned dictionaries (Song et al., 2026); recurrence alone does not validate their meaning.

4.2 Testing predictive and causal effects

A useful predictor need not identify a variable the source model uses. For a known target, SAE probes should be compared with supervised probes on the original activations, given matched information (Kantamneni et al., 2025; Makelov et al., 2025). For control, direct activation directions provide an alternative (Gurgurov et al., 2026). These comparisons test the benefit of the SAE representation for the chosen objective. A reusable feature vocabulary can still support exploratory discovery beyond a predefined prediction task (Peng et al., 2025).

Interventions test what changes when a feature is edited. Their effects must be separated from damage introduced by reconstruction. With error $e = h - (b + Dz)$, an error-preserving edit is

$$h' = b + Dz' + e = h + D(z' - z). \quad (2)$$

Replacing h with $b + Dz'$ also removes e, so the measured effect combines the feature edit with reconstruction error. Retaining e isolates the decoded change but does not ensure that the edited activation lies within the model's natural distribution. Null edits, matched-magnitude controls, naturally occurring replacement values, and off-target measurements test distinct alternatives. A controlled effect supports a causal claim about the manipulation; explaining the unmodified computation requires further evidence.